

COURSE TITLE
ENGINEERING ECONOMIC
ANALYSIS

Chapter 6 -RISK ANALYSIS

**Sensitivity Analysis, Decision Tree Analysis, Scenario
Analysis, Risk Adjusted MARR**

Learning Objective

After studying this lecture students will be able to learn:

- Concept of Sensitivity Analysis with example**
- Concept of Decision Tree Analysis with example**
- Concept of Scenario Analysis and Risk adjusted MARR**

SENSITIVITY ANALYSIS

**What are the Parameters
(variables) Used in
Engineering Economic
Analysis????**

SENSITIVITY ANALYSIS (Cont'd....)

They are :

- 1. Initial Cost (I)**
- 2. Annual Operating Cost (AOC)**
- 3. Salvage Value (S)**
- 4. Useful life (N)**
- 5. Interest rate (i)**
- 6. Inflation rate (f)**
- 7. Annual Revenues (A)**

SENSITIVITY ANALYSIS (Cont'd....)

- In economic analysis, all the estimate made of these parameters are **'best estimates'** made by experienced personnel i.e. the same value will occur during the implementation phase of the project.
- But in the real practice, the budgeting or estimate of these variables are **uncertain** i.e. there are variables which may not occur as expected.
- Sensitivity analysis tries to seek the solution what the outcome of the project will be if one of these **variables changes from the best estimates**.

SENSITIVITY ANALYSIS (Cont'd....)

Sensitivity analysis tries to answer the question of

‘ If the economic variables changes from expectation, what will the effect be on business, model system, or whatever being analyzed, and which variable are causing the largest deviations?’[1]

□ Sensitivity analysis determines the effect on the NPW of variations in the input variables (such as revenues, expenses, investment, salvage value etc.) used to estimate after tax cash flow. [2]

SENSITIVITY ANALYSIS (Cont'd....)

- “In Engineering economic studies, sensitivity analysis is a **general non probabilistic methodology**, to provide information about the potential impact of uncertainty in selected factor estimates”. [3]
- A Financial Sensitivity Analysis, also known as a **What-If analysis** or a **What-If simulation exercise**, is most commonly used by financial analysts to predict the outcome of a specific action when performed under certain conditions. [4]

SENSITIVITY ANALYSIS (Cont'd....)

For example:

- What if the investment increases by 20% of the initial investment, what will be the PW/IRR?**
- What if the net annual revenue decreases by 10%, what will be the changes in the PW/IRR?**
- What if the useful life increases by 15%, what will be the changes in PW/IRR ?**

Sensitivity analysis tries to answer these questions and is considered as what if analysis.

STEPS FOR SENSITIVITY ANALYSIS

- It begins with the **base case situation**, which is developed using the most likely values for each input. Develop the prime equation.
- Justify the project economically.
- Change the specific variable of interest by several specified percentages above and below the most – likely value, while holding other variables constant.
- Calculate a new PW/FW/AW/IRR/BCR for each of these values.

STEPS FOR SENSITIVITY ANALYSIS (Cont'd....)

- Present the results of a sensitivity analysis in the sensitivity graph.**
- The slope of the line shows how sensitive the NPW is to changes in each of the inputs.**
- The steeper the slope, the more the sensitive the NPW is to change in a particular variable**

SENSITIVITY ANALYSIS (Cont'd....)

Interpretation of Sensitivity Graph

- On a plot, there are two directions to measure uncertainty:**
 - On X-axis, the uncertainty in the input variable is measured.**
 - On Y-axis, the impact of that uncertainty on PW is measured.**
- The slope of the line shows how sensitive the PW is to change in each of the inputs. The steeper the slope, the more sensitive the PW is to a change in a particular variable.**
- The graph allows us to identify the crucial variables that most affect the final outcome**

SENSITIVITY ANALYSIS (Cont'd....)

□ A properly drawn spider plot shows the following:

- Limits of uncertainty for each cash flow element.**
- Impact of each cash flow element on the PW or IRR (each can be used as Y-axis)**
- Identification of each cash flow element that might change the recommendation.**

Example [3]

Perform the Sensitivity Analysis using the PW formulation over a range of $\pm 40\%$ changes in the estimates for –

- 1. Capital Investment (I)**
- 2. Annual net Cash flow (A)**
- 3. Market value (MV)**
- 4. Useful life (N)**

Based on these best estimates, plot a diagram that summarizes the sensitivity of present worth to percent deviation changes in each separate factor estimate when $MARR = 10\%$ per year

Example (Cont'd.....)

Capital Investment (I)	- \$ 11,500
Revenues / yr	\$ 5,000
Expenses / yr	- \$ 2,000
Market Value (MV)	\$ 1,000
Useful Life (N)	6 years

Example (Cont'd.....)

Step 1

□ Developing base case situation i.e. The PW of the project on the best estimate of the given factors is

Prime Equation

$$\begin{aligned}\text{PW (10\%)} &= - \$ 11,500 + \$ 3,000 (\text{P/A} , 10\% , 6) + \$ 1,000 (\text{P/F},10\%,6) \\ &= \$ 2,130 > 0 \text{ (economically accepted)}\end{aligned}$$

Step 2

When the capital Investment (I) varies with the increment of 10% up to $\pm 40\%$)

$$\begin{aligned}\text{PW (10\%)} &= - \$ 11,500 (1 \pm p\%) + \$ 3,000 (\text{P/A} , 10\% , 6) + \$ 1,000 \\ &\quad (\text{P/F},10\%,6)\end{aligned}$$

Example (Cont'd.....)

For + 10%,	PW (10%) = +\$980	For - 10%,	PW (10%) = +\$3280
For + 20%,	PW (10%) = - \$170	For - 20%,	PW (10%) = +\$4430
For + 30%,	PW (10%) = -\$1320	For - 30%,	PW (10%) = +\$5580
For + 40%,	PW (10%) = -\$2470	For - 40%,	PW (10%) = +\$6730

Step 3

When the Annual Cash Flow (A) varies with the increment of 10% up to $\pm 40\%$)

$$\text{PW (10\%)} = - \$ 11,500 + \$ 3,000 (1 \pm p\%) (P/A , 10\% , 6) + \$ 1,000 (P/F, 10\%, 6)$$

Example (Cont'd.....)

For + 10%, PW (10%) = +\$3,436 For - 10%, PW (10%) = +\$823.42
For + 20%, PW (10%) = +\$4743 For - 20%, PW (10%) = -\$483.16
For + 30%, PW (10%) = +\$6049 For - 30%, PW (10%) = -\$1789.74
For + 40%, PW (10%) = +\$7356 For - 40%, PW (10%) = -\$3096.32

Step 4

**When the Salvage Value (SV) varies with the increment of 10%
up to $\pm 40\%$**

$PW (10\%) = - \$ 11,500 + \$ 3,000 (P/A , 10\% , 6) + \$ 1,000 (1 \pm p\%) (P/F, 10\%, 6)$

Example (Cont'd.....)

For + 10%, PW (10%) = +\$2,186.44

For + 20%, PW (10%) = +\$2243

For + 30%, PW (10%) = +\$2299

For + 40%, PW (10%) = +\$2356.04

For - 10%, PW (10%) = +\$2073.56

For - 20%, PW (10%) = +\$2017.12

For - 30%, PW (10%) = +\$1960.68

For - 40%, PW (10%) = +\$1904.24

Example (Cont'd.....)

Step 5

When the useful life (N) varies with the increment of 10% up to $\pm 40\%$

$$\begin{aligned} \text{PW (10\%)} = & - \$ 11,500 + \$ 3,000 \{P/A, 10\%, 6 (1 \pm p\%) \} \\ & + \$ 1,000 \{P/F, 10\%, 6(1 \pm p\%) \} \end{aligned}$$

For + 10%, N=6.6 PW (10%) = +\$3040

For + 20%, N=7.2 PW (10%) = +\$3900

For + 30%, N=7.8 PW (10%) = +\$4710

For + 40%, N=8.4 PW (10%) = +\$5477

For - 10%, N=5.4 PW (10%) = +\$1167

For - 20%, N=4.8 PW (10%) = +\$147

For - 30%, N=4.2 PW (10%) = -\$933

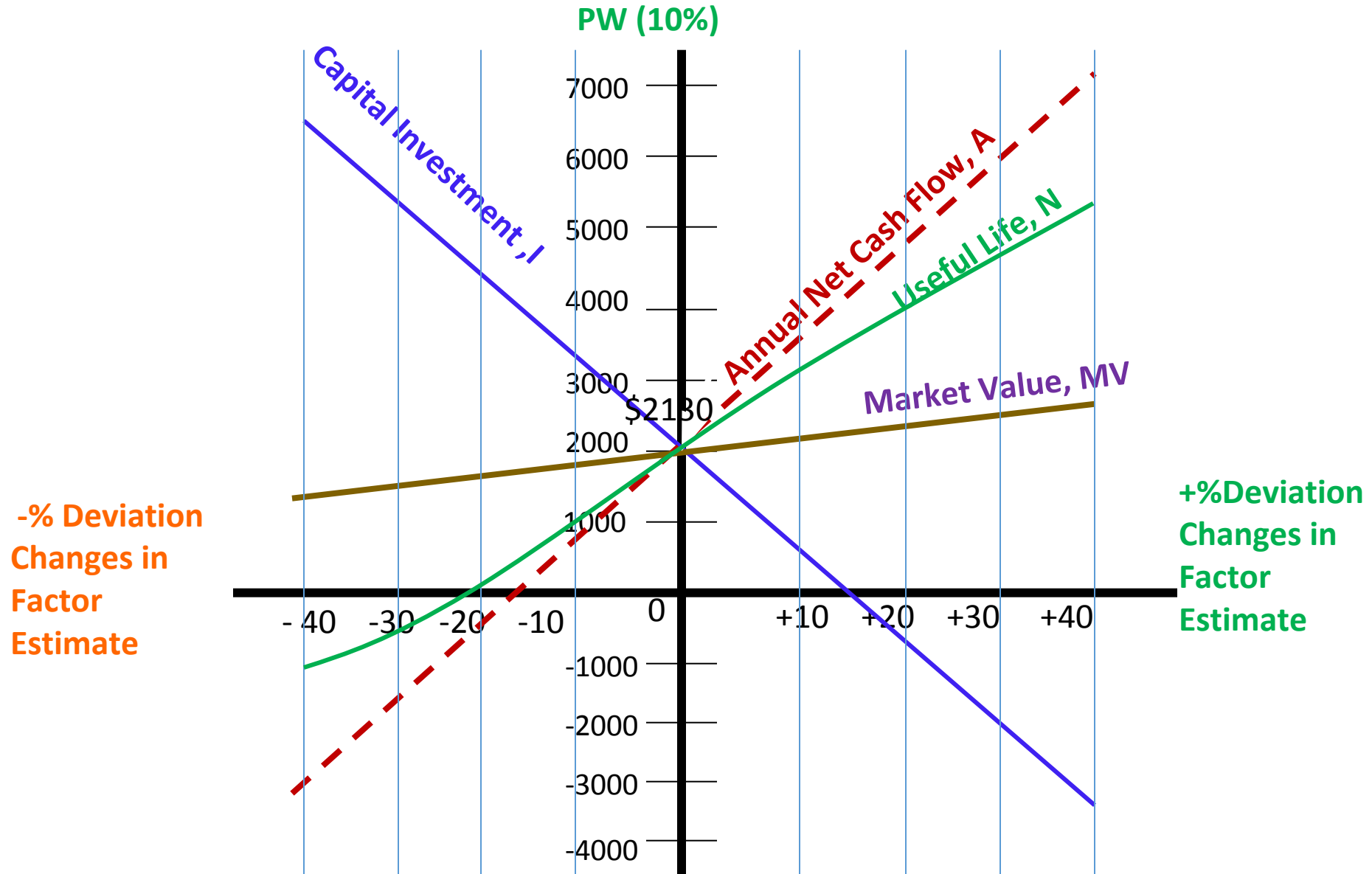
For - 40%, N=3.6 PW (10%) = -\$2077

Example (Cont'd.....)

Present Worth (PW)									
	- 40%	-30%	-20%	-10%	0%	+10%	+ 20%	+30%	+ 40%
I (\$)	6730	5,580	4430	3,280	2,130	980	- 170	-1320	- 2,470
A (\$)	-3,096.32	-1,789.74	- 483.16	823.42	2,130	3,436.58	4,743	6,049.74	7356.32
SV(\$)	1,904.24	1,960.68	2,017.12	2,073.56	2,130	2,186.44	2,243	2,299	2,356
N (\$)	- 2,786	-933	147	1167	2,130	3040	3,900	4,710	5,477

Plotting these values in the sensitivity graph

Step 6 Sensitivity Graph (Spider plot) Of Four Factors



Example (Cont'd.....)

Step 7 Revelations Of Spider plot

- Shows the sensitivity of the present worth to percent deviation changes in each factor's best estimate
- Other factors are assumed to remain at their best estimate values
- The relative degree of sensitivity of the present worth to each factor is indicated by the slope of the curves (the “steeper” the slope of a curve the more sensitive the present worth is to the factor)
- In this example:
Present worth is insensitive to MV and N
Present worth is sensitive to I, A,

Decision Tree Analysis

- ❑ Alternative evaluation may require a series of decisions where the outcome from one stage is important to the next stage of decision making.
- ❑ When each alternative is clearly defined and probability estimates can be made to account for risk, it is helpful to perform the evaluation using decision tree. [5]
- ❑ A decision tree is a flowchart that starts with one main idea and then branches out based on the consequences of your decisions.

Decision Tree Analysis

- Decision tree is a powerful means of facilitating the analysis of important problems, especially those that involve sequential decisions and variable outcomes over time. [3]
- It's called a “decision tree” because the model typically looks like a tree with branches. [6]
- In short, decision tree is the graphic representation of various alternative solutions that are available to solve a given problem, in order to determine the most effective courses of action. [7]

Decision Tree Analysis (Cont'd....)

A decision tree includes: [5]

- More than one stage of alternative selection.**
- Selection of an alternative at one stage that leads to another stage.**
- Expected results from a decision at each stage.**
- Probability estimates for each outcome.**
- Estimate of economic value (cost or revenue) for each outcome.**
- Measure of worth as the selection criterion such as $E(PW)$.**

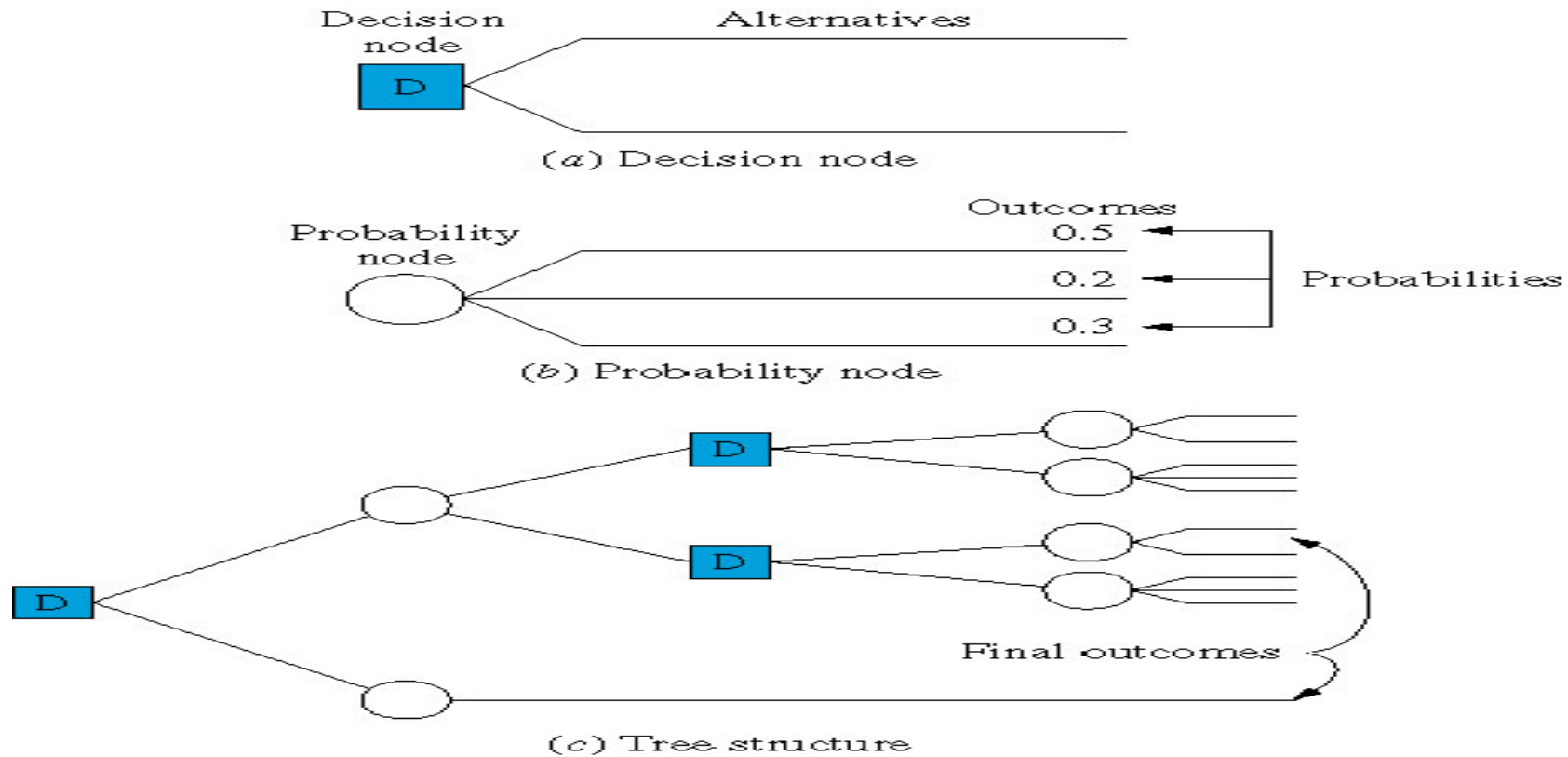
Decision Tree Analysis (Cont'd....)

Components of Decision Tree

The decision tree is constructed left to right and includes each possible decision and outcome.

- **Decision node:** A square represents a decision node for making decision by a decision maker.
- **Branch:** It is a line connecting nodes from the left to the right of the diagram.
- **Probability node (chance node) :** A circle represents probability node with the possible outcomes and estimated probabilities on the branches.

Decision Tree Analysis (Cont'd...)



Source: [5]

Decision Tree Analysis (Cont'd....)

For the Evaluation and Selection of the alternative, the following information is necessary. [5]

- The probability that is estimated must sum to 1.0 for each set of outcomes (branches) that results from decision.**
- Economic information for each decision alternative and possible outcome, such as initial investment and estimated cash flows.**

Decision Tree Analysis (Cont'd....)

Procedure for Solving Decision Tree Using PW analysis

- Start at the top right of the tree. Determine the PW value for each outcome branch.**
- Calculate the Expected value for each decision alternative**
- $E(\text{decision}) = \Sigma (\text{outcome estimate}) P(\text{outcome})$**
- At each decision node, select the best $E(\text{decision})$ value – minimum cost or maximum value (if both cost and revenues are estimated).**
- Continue moving to the left of the tree to the root decision in order to select the best alternative.**

Decision Tree Example

A company is planning for its plants by investing \$ 3,60,000. Estimates for efficiency of design goals, their probabilities and corresponding annual expenses saving are as follows:

Design goal met (%)	Probability	Annual expenses saving (\$)
85	0.35	40,000
60	0.50	33,000
50	0.15	26,000

The Company has another option to invest this money that earns 20% per year elsewhere. Based on the Expected PW as the decision criterion, determine whether the mechanizing is preferable or not? MARR = 12% and analysis period is 5 years.

Decision Tree Example (Cont'd....)

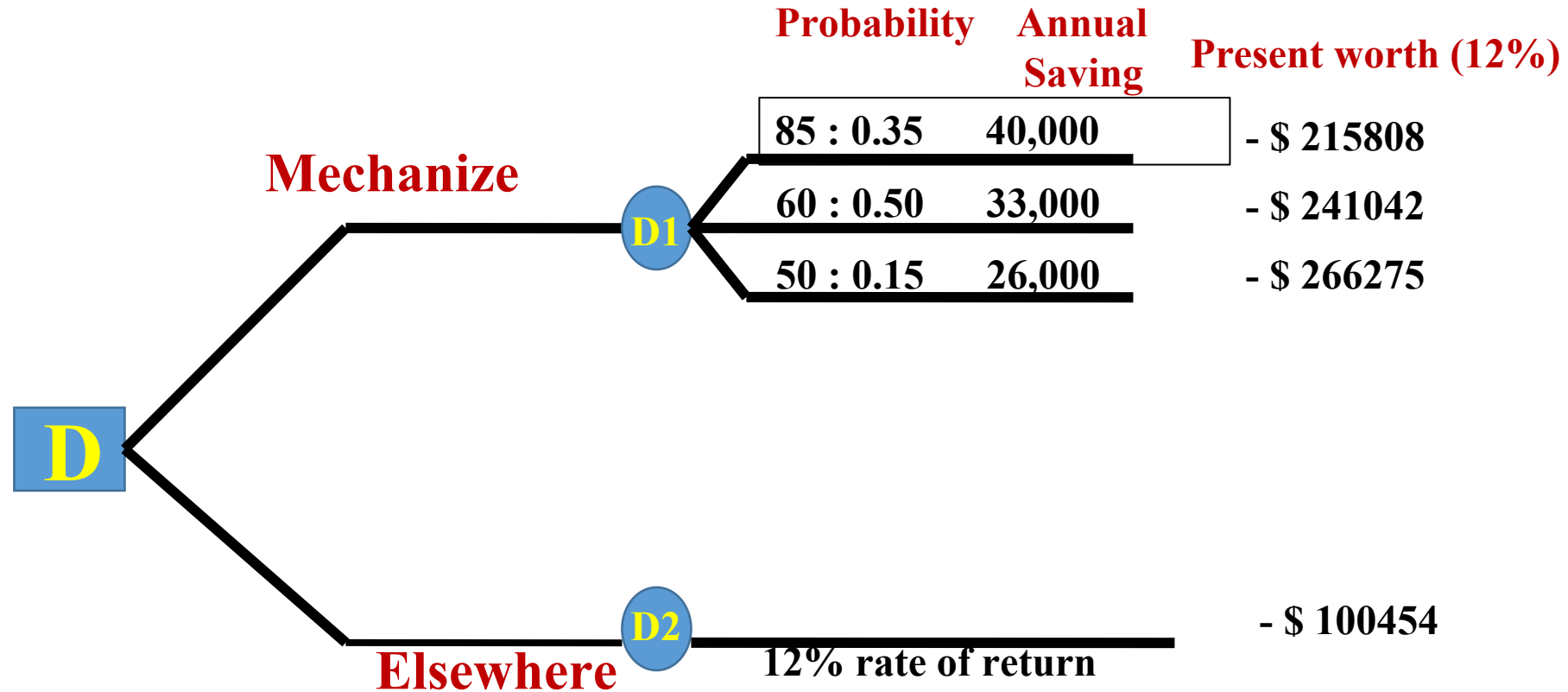
The PW corresponding to each efficiency goals are :

- **PW (12%)₈₅** = - 3,60,000 + 40,000 (P/A, 12%, 5) = - \$ 2,15,808
- **PW (12%)₆₀** = - 3,60,000 + 33,000 (P/A, 12%, 5) = - \$ 2,41,042
- **PW (12%)₅₀** = - 3,60,000 + 26,000 (P/A, 12%, 5) = - \$ 2,66,275

For Option 2 (Invest elsewhere that earns 20% per year)

- **Annual earning** = (0.2 * 3,60,000) = \$ 72,000
- **PW (12%)** = - 3,60,000 + 72,000 (P/A, 12%, 5) = - \$ 1,00,454

Decision Tree Example (Cont'd.....)



The single stage decision tree diagram

Decision Tree Example (Cont'd.....)

Expected PW

For option 1

$$\begin{aligned} E(PW) &= -215808(0.35) + (-241042)(0.50) + (-266275)(0.15) \\ &= -\$2,35,995 \end{aligned}$$

For Option 2

$$E(PW) = -\$100454$$

The analysis indicates that none of the options are preferable and even if we make the decision, the option 2 should be selected.

Scenario Analysis

- It is also called the Optimistic- pessimistic – most likely estimation.
- Sensitivity and break even analysis is useful to show the effect of one common variable and NPW.
- In reality there may be difficult to specify precisely the relationship between a particular variable and the NPW because there may be interdependence between the variable.
[8]
- Hence the relationship is further complicated by inter dependencies among the variables at a time.

Scenario Analysis (Cont'd...)

- **Scenario analysis is the technique that considers the sensitivity of NPW due to change in key variables at a time and the range of likely value of those variables.**
- **For example, the decision maker may examine three extreme cases i.e. best case scenario, worst case scenario and the most likely scenario.**
- **The NPWs under the worse and the best scenarios are then calculated and compared with the expected (or base case) NPW.**

Risk Adjust MARR

- Uncertainty causes factors inherent to engineering economy studies, such as cash flows and project life, to become random variables in the analysis.
- A widely used industrial practice for including some consideration of uncertainty is to increase the MARR when a project is thought to be relatively uncertain.
- Hence a procedure has emerged that employs risk adjusted interest rates.
- If two or more investments delivered the same return over a given time period, the one that has the lowest risk will have a better risk-adjusted return. [9]